

—— // ZPWR-DAW // ——

zpwr-daw

v0.1.0

MODULE REFERENCE

wire your own arranger from note-stream blocks — every track a patch graph

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MenkeTechnologies

—— MENKETECHNOLOGIES ——

Overview

zpw-r-daw started as a per-layer piano-roll pattern sequencer and grew into a two-view DAW: an Arrangement timeline (tracks, clips, sections, tempo/meter maps, markers, breakpoint automation) and a Session clip launcher (scenes, follow actions), plus MIDI/JSON export and an in-progress JUCE audio-clip layer.

It is a reusable component with two halves: a host-agnostic web UI (the FL-Studio-style canvas grid) and a C++ engine (pattern model, transport, step clock glue) reached directly from C++ and over a C ABI (FFI) from Rust. The frontend never knows whether it runs inside JUCE or Tauri – it calls an injected `nf` transport object. Playback runs on a native audio-thread step clock when the host wires it (minimise-proof) and falls back to a JS timer otherwise.

- One grid engine – one canvas renderer + one interaction model + a value model, bound to a `domain`; zero duplicated gesture code.
- N domains – notes (piano-roll), arranger (DAW arrangement), automation (ALS section-overrides), triggers (ztranslator).
- Two transport bridges – JUCE `WebBrowserComponent` native fns and Tauri `invoke`, behind one `nf` interface.
- Pure C++ `ClipEngine` – JUCE-free pattern model + swung step clock + event queue.
- C ABI + Rust bindings – `extern "C"` over `ClipEngine`; Rust hosts drive the same engine the plugins do.
- MIDI export – byte-identical C++ and JS Type-0 Standard MIDI File writers.

Generalized Grid Engine

The grid engine lives under `webui/grid/`: one renderer + one interaction model + one value model, bound to a `domain`. The domain supplies everything that differs (lanes, time-axis cells, value type, which gestures apply, serialize shape); the engine supplies everything shared (render loop, hit-test, all gestures, popover bulk-edit, persistence).

The merge detail that matters: `value.type` selects the edit-drag axis – `'length'` → horizontal right-edge note resize (notes), `'unit'` → vertical top-band value height (automation), `'bool'` → click toggle (triggers).

```
import { createGrid } from "../grid/index.js";
import { createNotesDomain } from "../grid/domains/notes.js";
import { createJuceBridge } from "../grid/transport/juce-bridge.js";

const { nf } = createJuceBridge({ prefix: uiPrefix }); // or createTauriBridge()
const grid = createGrid({
  canvas: document.getElementById("clip-grid"),
  domain: createNotesDomain({ getSteps, getPerBeat, layer: 0 }),
  store: localStorage,
  storageKey: "zfx_clip_notes_L0",
  onChange: (model) => nf("clipSeqPattern")(JSON.stringify(model.serialize())),
});
grid.setPlayhead(currentStep); // drive the playhead column from the transport
```

Grid Domains

A domain supplies `lanes()`, `cells()` over the time axis, a `value` spec (`type/min/max/step`), `capabilities` (which gestures apply), and `serialize/deserialize`. The bindings under `webui/grid/domains/`:

Domain	Lanes × cells	Value → serialize shape
<code>notes.js</code>	pitch lanes × step cells	note length → the CLIP piano-roll, <code>[[s,l,n,len,v]]</code> ClipSeq shape
<code>arranger.js</code>	tracks × bars	cells = clip ids → the DAW arrangement; double-click a clip drills into its notes
<code>automation.js</code>	macro lanes × 8-bar blocks	value 0..1 → the ALS section-overrides timeline, <code>{param:{bar:value}}</code> shape, resizable s
<code>triggers.js</code>	action lanes × time slots	value = bool → ztranslator (provisional, pending the ztranslator contract)
<code>autolanes.js</code>	MPE/CC lanes	per-block automation (Pitch Bend / CC74 / Channel Pressure as their real MIDI messages)
<code>launcher.js</code>	scene grid	Session clip launcher (scenes, follow actions)

Transport Bridges

The frontend isolates host calls behind `nf(name)(...args)`. Two adapters under `webui/grid/transport/` produce an `nf`, so the frontend never knows which host it runs in:

- `juce-bridge.js` – `nf` resolves to `Juce.getNativeFunction(prefix + name)`, i.e. the existing `clipSeq*` native fns from `registerClipSeqFns`. No backend change for the JUCE plugins.
- `tauri-bridge.js` – `nf` resolves to a wrapper over Tauri `invoke('clip_seq_' + name, {...})`. The Rust command handler forwards to the C ABI.

Pattern/transport JSON is identical across both bridges (defined in `ClipSeq.h`). Playhead readback: JUCE polls `clipSeqStep` on `rAF`; Tauri polls `invoke('clip_seq_step')` or listens for a Rust-emitted event.

Slot Sequencer

`webui/grid/sequencer.js` is a JS step clock that walks the time axis and fires every set cell per slot (swing-aware). It drives the triggers domain and is the JS-fallback clock for any host without a native engine. Native sequencer glue lives in `include/zpc/ClipSeq.h`: the `clipSeq*` host contract (`ClipSeqHooks`) plus a templated helper that registers the matching native functions on a JUCE `WebBrowserComponent` options builder.

```
#include <zpc/ClipSeq.h>

zpc::ClipSeqHooks clip;
clip.pattern    = [this] (const juce::String& json) { /* parse + stage pattern */ };
clip.transport = [this] (int steps, double bpm, double perBeat, float swing,
                          int swingUnit, bool loop, bool perLayer, int target) { /* ... */ };
clip.play       = [this] (bool on) { /* start / stop the processBlock step clock */ };
clip.step       = [this] { return currentStep; };

rootObject->setProperty ("hasClipSeq", clip.wired()); // advertise availability to the page
options = zpc::registerClipSeqFns (std::move (options), clip, pfx);
```

C++ Engine (ClipEngine)

`include/zpc/ClipEngine.h` is the pure C++ engine: a JUCE-free pattern model + swung step clock + event queue. Timing/length semantics are ported verbatim from the `clip.js` fallback sequencer, so the C++, the C ABI, and the JS clock all advance identically. It holds the pattern + transport, advances a step clock, exposes `currentStep()`, and queues per-step note on/off events – no `WebBrowserComponent` dependency.

The note-stream module pack (`include/zpc/midi/`, `src/midi/`) – Chord / Arp / Scale / Euclidean / Game-of-Life / Brian-Brain / Langton plus the chord & scale engines – are `ModuleInfoT<NoteStream>` modules templated on the signal-agnostic graph core from `zpw-r-patch-core` (vendored under `libs/zpw-r-patch-core`; only `juce_core` beyond that).

C ABI + Rust Bindings

The pure C++ `ClipEngine` is exposed to Rust over an `extern "C"` surface (`include/zpc/capi/clip_engine.h`, `src/capi/clip_engine.cpp`) so Audio-Haxor / `ztranslator` drive the same engine the JUCE plugins drive. The `-sys` crate compiles the JUCE-free `clip_engine.cpp` directly with `cc` – no CMake or JUCE needed for the Rust path. The `safe` wrapper crate sits on top.

```
use zpw_clip_engine::ClipEngine;
let mut e = ClipEngine::new();
e.set_pattern(r#"["s":0,"l":0,"n":60,"len":2,"v":100}]"#);
e.set_transport(4, 120.0, 4.0, 0.0, 1, /*loop*/ true, /*per_layer*/ false, /*target*/ 0);
e.play(true);
e.advance(0.125); // one step at 120 bpm / 1/16
for ev in e.poll_events() { /* route note on/off to the synth */ }
```

C/C++ consumers can instead link the CMake target `zpw::clip_engine_capi` (static) or build `-DZPC_BUILD_CAPI_SHARED=ON` for a shared library.

Arrangement & Session

Two views over the same engine:

- Arrangement – a timeline of tracks, clips, sections, tempo/meter maps, markers and breakpoint automation. The `arranger` domain renders tracks × bars with cells = clip ids; double-click a clip drills into its notes.
- Session – a clip launcher with scenes and follow actions (`launcher.js`).

The detailed per-feature coverage across Ableton Live, FL Studio, Logic, Cubase/Studio One, Bitwig and Pro Tools – every PORTED row cited to its `file:line` – is in the [DAW Arranger Port Report](#).

MIDI / JSON Export

Any pattern renders to a Type-0 Standard MIDI File – the universal DAW interchange format – so the sequencer's output drops straight into Ableton, FL, Logic, Reaper, etc. One step = one grid cell; `perBeat` is steps-per-quarter (the `clip.js` note division); layer → MIDI channel; `repeats` renders the pattern back-to-back N times. The C++ (`include/zpc/MidiFile.h`) and JS (`webui/grid/export/midi.js`) writers emit byte-identical files, verified by round-trip parsing in the test suites.

```
// Rust / native
e.write_midi_file("clip.mid", /*repeats*/ 4)?;    // or e.render_midi(4) -> Vec<u8>

// Web (grid frontend)
downloadMidi(patternToMidi(grid.serialize(), { bpm, perBeat, steps, repeats: 4 })), "clip.mid");

// C ABI
zpc_clip_render_midi(engine, repeats, &data, &len);    // free with zpc_bytes_free
```

Build / Layout

Point `ZPC_JUCE_DIR` at a JUCE checkout. Standalone builds compile the two graph-core translation units from the submodule directly, so no separate `zpwr_patch_core` lib is required; when embedded in a host that already defines `zpwr_patch_core`, this links that target instead.

```
# standalone test
cmake -S . -B build -DZPC_JUCE_DIR=/path/to/JUCE
cmake --build build --target MidiModulesTest
./build/MidiModulesTest_artefacts/*/MidiModulesTest

# pure-logic grid tests (headless)
node --test webui/grid/tests/grid.test.mjs

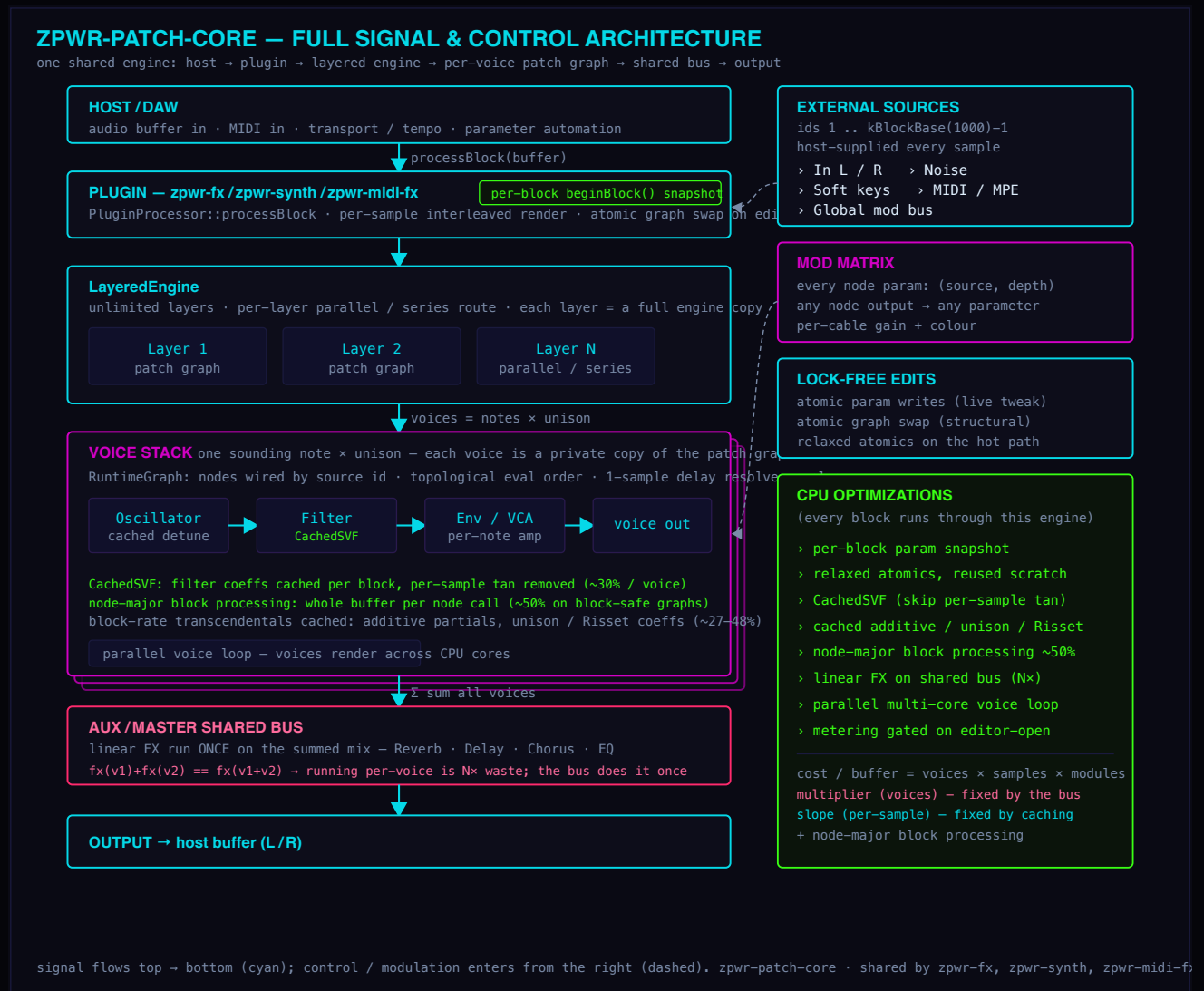
# Rust bindings
cargo test --manifest-path bindings/rust/zpwr-clip-engine/Cargo.toml
```

Path	Role
<code>webui/grid/</code>	Generalized canvas grid engine (renderer + interaction model + value model)
<code>webui/grid/domains/</code>	Domain bindings – notes, arranger, automation, triggers, autolanes, launcher
<code>webui/grid/transport/</code>	JUCE and Tauri transport bridges (one <code>nf</code> interface)
<code>webui/grid/sequencer.js</code>	Swing-aware JS step clock; JS-fallback clock
<code>webui/clip/</code>	Legacy DOM piano-roll (<code>clip.js</code>); stays live until consumers cut over
<code>include/zpc/ClipEngine.h</code>	Pure C++ engine – pattern model + swung step clock + event queue
<code>include/zpc/ClipSeq.h</code>	Native sequencer glue – <code>ClipSeqHooks</code> + <code>registerClipSeqFns</code>
<code>include/zpc/MidiFile.h</code>	Type-0 Standard MIDI File writer (C++)
<code>include/zpc/capi/clip_engine.h</code>	C ABI (<code>extern "C"</code>) over <code>ClipEngine</code>
<code>src/capi/clip_engine.cpp</code>	C ABI implementation
<code>include/zpc/midi/, src/midi/</code>	Note-stream module pack (Chord / Arp / Scale / Euclidean / cellular automata)
<code>bindings/rust/</code>	<code>zpwr-clip-engine-sys</code> (raw decls, builds the C ABI via <code>cc</code>) + <code>zpwr-clip-engine</code> (safe wrapper)
<code>libs/zpwr-patch-core</code>	Vendored signal-agnostic graph core submodule

© MenkeTechnologies – the DAW arranger engine of the MenkeTechnologies audio stack, shared across `zpwr-synth` · `zpwr-midi-fx` · `zpwr-fx` · Audio-Haxor · `ztranslator`.
References: [Engineering Report](#) · [DAW Arranger Port Report](#) · [Source](#).

Architecture

ZPWR-DAW runs on the shared zpwr-patch-core engine. The full signal and control path – from the host buffer, through the layered engine and the per-track patch graph, onto the shared bus, and back out – is shown below.



Signal Sources

External sources you can cable into any block input or modulation route.

1 MIDI In 3 Random Mod Wheel Pitch Bend Aftertouch Velocity Expression Sustain MPE Bend MPE Pressure MPE Slide

Modulation

25 modules

Generative 1 input

Probabilistic melodic random walk synced to a rate.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	—	—
Range	1 - 12	3	st
Low	0 - 127	48	—
High	0 - 127	72	—
Probability	0 - 1	0.7	—

MidiLFO 1 input

Low-frequency modulation source (sine/tri/saw/square) on the scalar control output.

Param	Range	Default	Unit
Rate	0.01 - 20	1	Hz
Shape	Sine · Triangle · Saw · Square	—	—
Depth	0 - 1	1	—

MidiEnv 1 input

ADSR envelope triggered by the note gate, on the scalar control output.

Param	Range	Default	Unit
Attack	1 - 5000	5	ms
Decay	1 - 5000	200	ms
Sustain	0 - 1	0.7	—
Release	1 - 5000	300	ms

MidiRandom 1 input

Generates random notes within a pitch range at a synced rate.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	—	—
Low Note	0 - 127	48	—
High Note	0 - 127	72	—
Gate	0 - 2	0.5	—
Velocity	1 - 127	100	—

MidiSampleHold 1 input

Samples and holds the incoming control value on each note.

MidiSlew 1 input

Smooths (glides) the control value over a set time.

Param	Range	Default	Unit
Time	0 - 2000	100	ms

RandOctave 1 input

Randomly shifts notes by up to +/- Range octaves with a chance.

Param	Range	Default	Unit
Range	1 - 3	1	-
Chance	0 - 1	1	-

CC LFO 1 input

Continuous CC LFO generator (sine/triangle/saw/square) emitted on a chosen CC at a free Rate - automates a synth parameter over MIDI, where MidiLFO drives only the internal mod-matrix scalar.

Param	Range	Default	Unit
CC	0 - 127	1	-
Rate	0.01 - 20	1	Hz
Shape	Sine · Triangle · Saw · Square		-
Depth	0 - 1	1	-

CCSeq 1 input

Tempo-synced step CC sequencer: one CC value per synced step cycling an 8-step shape (overridable via the node's config list) - a modulation sequencer in the CC domain, where StepSeqMidi sequences notes.

Param	Range	Default	Unit
CC	0 - 127	1	-
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		-

VelToCC 1 input

Turns each note's velocity into a CC value emitted just before the note, so a synth's filter or level opens with how hard you played - dynamics routed to expression, where Note2CC sends pitch.

Param	Range	Default	Unit
CC	0 - 127	11	-

CCGlide 1 input

Portamento for a CC: smooths the watched CC toward each newly received value over Time, emitting interpolated CC each block - de-zippers a stepped controller, where MidiSlew only glides the internal scalar.

Param	Range	Default	Unit
CC	0 - 127	1	-
Time	1 - 2000	120	ms

PressSwell 1 input

Generates a channel-pressure (aftertouch) swell that rises to Max over Rise while any note is held, snapping back to zero on release - an automatic crescendo for pads and strings, where ATProc only transforms incoming aftertouch.

Param	Range	Default	Unit
Rise	1 - 5000	800	ms
Max	0 - 1	1	-

PitchDrift 1 input

Nudges each note's pitch by a small random detune up to +/-Cents (per-note, polyphonic) - the gentle oscillator instability of analog gear, applied to pitch where Humanize only scatters timing and velocity.

Param	Range	Default	Unit
Cents	0 - 50	8	c

MicrotuneSpread 1 input

Microtune spread: gives every note a small random pitch detune up to +/-Cents via its microtuning offset, loosening rigid tuning into a chorused, slightly-out ensemble shimmer.

Param	Range	Default	Unit
Cents	0 - 50	10	c

QuarterToneShift 1 input

Quarter-tone shift: detunes every note by up to a quarter-tone via its microtuning offset, sliding the whole performance off the 12-tone grid into 24-tone territory without changing the note numbers.

Param	Range	Default	Unit
Amount	-1 - 1	0.5	-

VelToDetune 1 input

Velocity-to-detune: maps how hard each note is played to a microtuning detune, so dynamics bend pitch slightly sharp or flat - the pitch-pulls-with-force behaviour of acoustic instruments and analog drift.

Param	Range	Default	Unit
Cents	-50 - 50	15	c

GamelanDetune 1 input

Gamelan detune: tunes alternate keys slightly apart so paired notes shimmer with the slow acoustic beating of a gamelan's deliberately-mistuned instrument pairs (ombak) - an ensemble-wide chorused waver.

Param	Range	Default	Unit
Cents	0 - 50	15	c

GoldenAngleDetune 1 input

Golden-angle detune: scatters each successive note's microtuning by the golden angle, the maximally-even irrational spacing, so repeated notes never land on the same detune - an organic, never-repeating pitch shimmer.

Param	Range	Default	Unit
Cents	0 - 50	12	c

StretchTuning 1 input

Stretch tuning: applies a Railsback-style octave stretch, tuning notes progressively sharp as they rise and flat as they fall, the psychoacoustic tuning real pianos use so octaves sound in tune to the ear.

Param	Range	Default	Unit
Cents	-30 - 30	8	c

NEdoRetune 1 input

N-EDO retune: detunes every note from 12-tone equal temperament to the nearest step of an N-equal-divisions-of-the-octave scale, opening up 19, 24, 31 or any microtonal grid without changing the played note numbers.

Param	Range	Default	Unit
Divisions	5 - 53	19	-

HarmonicSeriesSnap 1 input

Harmonic-series snap: detunes each note to the nearest member of the natural harmonic series above a root, replacing tempered intervals with the pure, beatless ratios of just intonation as a brass or didgeridoo would sound them.

Param	Range	Default	Unit
Root	0 - 60	24	-

WerckmeisterTune 1 input

Werckmeister tune: retunes to Werckmeister III, the famous 1691 well-temperament whose unequal fifths give each key its own colour (the 'well-tempered' tuning of Bach's era) - subtle, key-dependent detuning from equal temperament.

Param	Range	Default	Unit
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	-	-

PythagoreanTune 1 input

Pythagorean tune: retunes to 3-limit Pythagorean intonation built from a spiral of pure fifths, giving brilliant, wide major thirds and the bright, slightly-sharp leading tones of medieval and string-ensemble tuning.

Param	Range	Default	Unit
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	-	-

BohlenPierceTune 1 input

Bohlen-Pierce tune: retunes the keyboard to the Bohlen-Pierce scale, which divides not the octave but the 3:1 'tritave' into 13 equal steps, giving an alien but surprisingly consonant tuning built on odd-harmonic ratios.

Param	Range	Default	Unit
Root	0 - 127	60	-

CarlosAlphaTune 1 input

Carlos Alpha tune: retunes to Wendy Carlos's Alpha scale of 78-cent equal steps, a non-octave temperament engineered to nail pure major thirds and fifths at the cost of ever repeating at the octave - a shimmering, otherworldly intonation.

Param	Range	Default	Unit
Root	0 - 127	60	-

Dynamics

63 modules

Velocity 1 input

Scales and offsets note velocity, with a modulation amount.

Param	Range	Default	Unit
Scale	0 - 2	1	-
Offset	-64 - 64	0	-
Mod Amt	-64 - 64	0	-

VelCurve 1 input

Reshapes velocity through a power curve.

Param	Range	Default	Unit
Curve	0.2 - 5	1	-

Humanize 1 input

Adds random timing and velocity jitter for a human feel.

Param	Range	Default	Unit
Timing	0 - 50	8	ms
Velocity	0 - 1	0.3	-

Accent 1 input

Boosts velocity on every Nth note.

Param	Range	Default	Unit
Every	1 - 16	4	-
Amount	0 - 1	0.5	-

VelClip 1 input

Clamps velocity between Min and Max.

Param	Range	Default	Unit
Min	1 - 127	1	-
Max	1 - 127	127	-

Ramp 1 input

Velocity ramp over a run of notes, ascending or descending.

Param	Range	Default	Unit
Length	2 - 16	8	-
Depth	0 - 1	0.6	-
Direction	Up · Down		-

VelCompress 1 input

Velocity compressor: velocities above Threshold are pulled toward it by Ratio (1 = off, 8 = hard), narrowing the dynamic range smoothly instead of hard-clamping like VelClip.

Param	Range	Default	Unit
Threshold	1 - 127	64	-
Ratio	1 - 8	2	-

VelInvert 1 input

Flips the velocity scale around its midpoint (soft <-> loud) by Amount, so your accents become ghost notes and the ghosts become accents - a velocity mirror, blendable to taste.

Param	Range	Default	Unit
Amount	0 - 1	1	-

VelKey 1 input

Keyboard velocity tracking: scales each note's velocity by how far its pitch sits from a Center note, Tilt setting how strongly and which way - higher notes louder (or softer), the dynamic key-tracking of real instruments.

Param	Range	Default	Unit
Center	0 - 127	60	-
Tilt	-1 - 1	0.5	-

Vellength 1 input

Note duration scales with velocity: hard notes ring for Max ms, soft notes for Min (set Max < Min to make accents staccato) - the played articulation where dynamics shape note length.

Param	Range	Default	Unit
Min	1 - 2000	60	ms
Max	1 - 2000	500	ms

GoldenVel 1 input

Golden velocity: replaces each note's velocity with a value from the golden-ratio low-discrepancy sequence, which fills the dynamic range far more evenly than random, so accents never clump; Depth blends it against the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

RudinShapiroAccent 1 input

Rudin-Shapiro accent: pushes each note's velocity up or down by the +/-1 Rudin-Shapiro sequence of its count, an aperiodic accent pattern with a famously flat (white-like) spectrum - structured stress that never repeats predictably.

Param	Range	Default	Unit
Amount	0 - 63	24	-

FibVelocity 1 input

Fibonacci velocity: drives note velocity from the Fibonacci sequence taken modulo 128, climbing then wrapping in the golden cadence; Depth blends the pattern against the incoming velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

VelDrift 1 input

Velocity drift: each note's velocity takes one step of a bounded random walk, so dynamics wander smoothly and correlatedly over time instead of jumping independently like Humanize - the slow swell and ebb of a tiring player.

Param	Range	Default	Unit
Step	0 - 1	0.3	-

VelGate 1 input

Velocity gate: drops any note whose velocity falls outside the Min..Max window, so only soft (or only hard) hits pass - a dynamics-dependent filter for splitting ghosted from accented playing.

Param	Range	Default	Unit
Min	1 - 127	1	-
Max	1 - 127	127	-

CrescendoLoop 1 input

Crescendo loop: velocity climbs from Min to Max across Length notes then snaps back and repeats, an automatic sawtooth dynamic swell driven by the note count rather than the clock.

Param	Range	Default	Unit
Length	2 - 32	8	-
Min	1 - 127	30	-
Max	1 - 127	120	-

VelByPitch 1 input

Velocity by pitch: sets each note's velocity from its keyboard position (a Slope per semitone around middle C plus a Base), so high notes can ring louder or softer - keyboard-position dynamics.

Param	Range	Default	Unit
Slope	-3 - 3	1	-
Base	1 - 127	80	-

ZipfVelocity 1 input

Zipf velocity: draws each note's velocity from a Zipf (1/rank) distribution, so a few dynamic levels dominate and loud accents are rare - the power-law dynamics of natural performance; Depth blends it against the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

PrimeVelocity 1 input

Prime velocity: drives velocity from the gaps between successive prime numbers, an irregular yet deterministic accent sequence that never settles into an obvious loop; Depth blends it with the incoming velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

VelStep 1 input

Velocity step: quantizes velocity to a small number of levels (a dynamics bit-crush), flattening expressive playing into stepped terraces from gentle 8-level to brutal 2-level.

Param	Range	Default	Unit
Levels	2 - 16	4	-

AccentPattern 1 input

Accent pattern: boosts velocity on the on-steps of a 16-step accent bitmask and trims it elsewhere, stamping a repeating dynamic groove onto an even stream of notes.

Param	Range	Default	Unit
Pattern	0 - 65535	34953	-
Amount	0 - 63	24	-

ThueMorseVelocity 1 input

Thue-Morse velocity: alternates two velocity levels by the cube-free Thue-Morse parity of the note count, accenting in a self-similar, never-quite-repeating pattern instead of a simple every-other.

Param	Range	Default	Unit
Low	1 - 127	50	-
High	1 - 127	110	-

CollatzVelocity 1 input

Collatz velocity: maps each note's velocity to the number of $3n+1$ steps its count takes to reach 1, an erratic deterministic accent sequence from the unsolved conjecture; Depth blends against the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

DigitSumVelocity 1 input

Digit-sum velocity: derives velocity from the decimal digit-sum of the note count, a self-similar sawtooth-of-sawtooths accent that resets every power of ten; Depth blends it with the incoming velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

HarmonicVelocity 1 input

Harmonic velocity: shapes velocity along a $1/(k+1)$ harmonic-decay curve over a repeating length-note cycle, so each phrase opens strong and tapers - an automatic accent-and-decay groove.

Param	Range	Default	Unit
Length	2 - 16	4	-
Base	1 - 127	110	-

DivisorCountVelocity 1 input

Divisor-count velocity: makes notes louder on counts with many divisors, so highly-composite ordinals (12, 24, 36...) land as accents and primes stay quiet; Depth blends against the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

TotientVelocity 1 input

Totient velocity: drives velocity from Euler's totient ratio $\phi(n)/n$ of the count, dipping on numbers rich in small prime factors and peaking on primes - a number-theoretic accent contour; Depth blends with the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

OmegaVelocity 1 input

Omega velocity: sets velocity from the number of prime factors (with multiplicity) of the count, so smooth primes play soft and factor-heavy numbers play hard; Depth blends it with the incoming velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

PitchClassVelocity 1 input

Pitch-class velocity: accents every note of a chosen pitch class and softens the rest, so a tonic or any scale degree rings out across the whole keyboard - a harmonic spotlight on one note name.

Param	Range	Default	Unit
Class	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	-	-
Amount	0 - 63	24	-

IntervalVelocity 1 input

Interval velocity: sets each note's velocity from the size of its leap from the previous note, so wide jumps hit hard and stepwise motion stays gentle - dynamics that follow the melodic contour; Depth blends with the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

RepeatDecay 1 input

Repeat decay: each immediate repetition of the same note plays quieter by a decay factor, taming machine-gun retriggers into a natural fade and resetting the moment the pitch changes.

Param	Range	Default	Unit
Decay	0.3 - 0.99	0.8	-

BinaryWeightVelocity 1 input

Binary-weight velocity: drives velocity from the number of 1-bits (Hamming weight) in the note count, a self-similar accent pattern that pulses with the binary structure of the ordinal; Depth blends it with the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

RulerVelocity 1 input

Ruler velocity: accents notes by the ruler sequence (the power of two dividing the count: 0,1,0,2,0,1,0,3,...), so every other note is a light tick and the downbeats of each binary level land harder - a self-similar metric accent.

Param	Range	Default	Unit
Depth	0 - 1	1	-

SternBrocotVelocity 1 input

Stern-Brocot velocity: shapes velocity from Stern's diatomic (fusc) sequence, the fractal numerators of the Stern-Brocot tree of rationals, giving a self-similar sawtooth-of-sawtooths accent; Depth blends with the incoming velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

VelocityLFO 1 input

Velocity LFO: sweeps velocity up and down with a slow sine over a set number of notes, an automatic dynamic swell-and-fade that breathes life into a flat sequence; Depth sets how deep the swell.

Param	Range	Default	Unit
Period	2 - 64	16	-
Depth	0 - 63	30	-

SwingVelocity 1 input

Swing velocity: accents on-beat notes and softens off-beat ones, applying the dynamic side of a swing/shuffle feel (the loud-soft alternation) independent of timing - instant groove from even input.

Param	Range	Default	Unit
Amount	0 - 63	20	-

GoldenSectionAccent 1 input

Golden-section accent: places a strong accent at the start and at the golden-ratio point (~0.618) of a repeating cycle, an aesthetically-balanced asymmetric stress pattern instead of an even backbeat.

Param	Range	Default	Unit
Length	3 - 32	8	-
Amount	0 - 63	28	-

ZeckendorfVelocity 1 input

Zeckendorf velocity: accents each note by how many Fibonacci numbers it takes to sum to the note count (its Zeckendorf representation length), a number-theoretic accent that grows in golden steps; Depth blends with the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

VelocitySmooth 1 input

Velocity smooth: glides each note's velocity toward a running average of the recent ones, ironing out wild dynamic jumps into an even, controlled performance; Amount sets how strongly it pulls toward the average.

Param	Range	Default	Unit
Amount	0 - 1	0.6	-

MarkovVelocity 1 input

Markov velocity: walks the velocity through a near-neighbour Markov chain of levels, so dynamics drift coherently (mostly small changes, occasional leaps) rather than jumping randomly - a stochastic but musical accent contour.

Param	Range	Default	Unit
Levels	3 - 12	6	-
Wander	0 - 1	0.4	-

EuclideanVelocity 1 input

Euclidean velocity: accents the notes that fall on the onsets of a Euclidean rhythm of Pulses-in-Steps and softens the rest, stamping a maximally-even world-rhythm groove onto the dynamics.

Param	Range	Default	Unit
Pulses	1 - 16	5	-
Steps	1 - 16	8	-
Amount	0 - 63	28	-

VelGateHysteresis 1 input

Velocity Schmitt gate: opens once a note exceeds the High velocity and stays open until one drops below Low, so soft passages between accents either all pass or all mute - hysteretic dynamic gating that avoids chattering at the threshold.

Param	Range	Default	Unit
High	1 - 127	90	-
Low	1 - 127	50	-

KolakoskiVelocity 1 input

Kolakoski velocity: alternates two velocity levels following the self-describing Kolakoski sequence (whose run-lengths are the sequence itself), giving a hypnotically self-similar yet non-repeating accent pattern.

Param	Range	Default	Unit
Low	1 - 127	50	-
High	1 - 127	110	-

MoebiusVelocity 1 input

Moebius velocity: accents notes by the Moebius function of the count - loud on squarefree numbers with an even number of prime factors, soft on odd, and mid-level on square-divisible ones; Depth blends with the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

PadovanVelocity 1 input

Padovan velocity: steps velocity through the Padovan sequence ($P(n)=P(n-2)+P(n-3)$, the plastic-number recurrence) taken modulo a range, a slow, gently-rolling accent contour; Depth blends with the incoming velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

LucasVelocity 1 input

Lucas velocity: steps velocity through the Lucas numbers (the Fibonacci companion 2,1,3,4,7,11,...) taken modulo a range, a golden-cadence accent contour; Depth blends with the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

GcdVelocity 1 input

GCD velocity: sets velocity from the greatest common divisor of the note count and a modulus, so counts sharing big factors with the modulus hit harder - a quietly-periodic, factor-driven accent; Depth blends with the played velocity.

Param	Range	Default	Unit
Modulus	2 - 24	12	-
Depth	0 - 1	1	-

PrimeCountVelocity 1 input

Prime-count velocity: drives velocity from the prime-counting function $\pi(\text{count})$ - how many primes are at or below the note count - taken modulo a span, a slowly-climbing-then-wrapping accent rooted in the distribution of primes.

Param	Range	Default	Unit
Depth	0 - 1	1	-

MelodicAccent 1 input

Melodic accent: accents notes that step up from the previous one and softens those that step down, so the dynamics follow the rise and fall of the melodic contour - phrasing that breathes with the line.

Param	Range	Default	Unit
Amount	0 - 63	20	-

VelocityFloor 1 input

Velocity floor: lifts any note played softer than the floor up to it, guaranteeing every note speaks at a minimum level - a reverse noise gate for taming ghost notes and uneven controllers.

Param	Range	Default	Unit
Floor	1 - 127	40	-

VelocityExpand 1 input

Velocity expand: pushes velocities away from the mid-level (the opposite of compression), so soft notes get softer and loud notes louder - widening the dynamic range of a flatly-played or over-quantized part.

Param	Range	Default	Unit
Ratio	1 - 3	1.5	-

AccentEveryN 1 input

Accent every N: boosts the velocity of every Nth note and slightly ducks the rest, stamping a steady metric downbeat onto an even stream - the simplest way to imply a time signature in the dynamics.

Param	Range	Default	Unit
N	2 - 16	4	-
Amount	0 - 63	28	-

GhostNoteInject 1 input

Ghost-note inject: randomly demotes a fraction of notes to a soft 'ghost' level, scattering the quiet in-between hits that give drum and bass parts their human, shuffling feel.

Param	Range	Default	Unit
Probability	0 - 1	0.3	-
GhostVel	1 - 80	30	-

DigitalRootVelocity 1 input

Digital-root velocity: accents each note by the digital root (the repeated digit-sum, 1..9) of its count, a perfectly periodic nine-step accent staircase from elementary number theory; Depth blends with the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

AbundancyVelocity 1 input

Abundancy velocity: drives velocity from the abundancy index $\sigma(n)/n$ of the count - near 1 for primes, climbing past 2 for perfect and abundant numbers - so divisor-rich counts hit hardest; Depth blends with the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

PrimeFactorVelocity 1 input

Prime-factor velocity: sets velocity from the largest prime factor of the count (log-scaled), so prime counts ring loud and smooth, small-factored counts stay soft; Depth blends with the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

SquareDistanceVelocity 1 input

Square-distance velocity: accents notes by how close the count sits to a perfect square - loud right on the squares, fading in the gaps - a pulsing accent that widens as the squares spread apart; Depth blends with the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

VelocityRandomScale 1 input

Velocity random scale: multiplies each note's velocity by an independent random gain, scattering the dynamics for a more human, less machine-perfect feel (timing untouched, unlike Humanize); Amount sets the scatter width.

Param	Range	Default	Unit
Amount	0 - 1	0.3	-

HappyVelocity 1 input

Happy velocity: accents each note by how its count behaves under the happy-number process (sum of squared digits) - loud and bright if it reaches 1, darker if it falls into the sad cycle; Depth blends with the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

KaprekarVelocity 1 input

Kaprekar velocity: drives velocity from how many digit-sort-and-subtract steps the count takes to reach Kaprekar's constant 6174, an erratic 0-7 accent from a famous digit routine; Depth blends with the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

CatalanVelocity 1 input

Catalan velocity: steps velocity through the Catalan numbers (1,1,2,5,14,42,...) taken modulo a range, the combinatorial sequence counting balanced brackets and binary trees; Depth blends with the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

DigitReverseVelocity 1 input

Digit-reverse velocity: drives velocity from the count read backwards (123 becomes 321), an erratic accent that scrambles the steady climb of the counter into a jumpy pattern; Depth blends with the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

PersistenceVelocity 1 input

Persistence velocity: accents each note by the multiplicative-persistence depth of its count (how many digit-product steps reach a single digit), a sparse, spiky accent rooted in a famous open digit problem; Depth blends with the played velocity.

Param	Range	Default	Unit
Depth	0 - 1	1	-

Control

5 modules

CCMap 1 input

Remaps a CC number and scales/offsets its value.

Param	Range	Default	Unit
From	-1 - 127	1	-
To	0 - 127	11	-
Scale	0 - 2	1	-
Offset	-1 - 1	0	-

Note2CC 1 input

Emits a CC from each note (pitch or velocity).

Param	Range	Default	Unit
CC	0 - 127	1	-
Source	Pitch · Velocity		-

CC2Note 1 input

A watched CC crossing a threshold triggers a note.

Param	Range	Default	Unit
Watch	0 - 127	1	-
Note	0 - 127	60	-
Threshold	0 - 1	0.5	-

BendProc 1 input

Scales and/or inverts incoming pitch-bend.

Param	Range	Default	Unit
Scale	0 - 2	1	-
Invert	0 - 1	0	-

ATProc 1 input

Scales aftertouch, optionally converting it to a CC.

Param	Range	Default	Unit
Scale	0 - 2	1	-
To CC	-1 - 127	-1	-

Expression

3 modules

Scoop 1 input

Bends up into each struck note from Depth below, resolving to pitch over Time - the vocal/brass scoop, a pitch-bend ramp that ends at centre so no bend is left hanging.

Param	Range	Default	Unit
Depth	0 - 1	0.3	-
Time	1 - 1000	80	ms

Fall 1 input

Jazz fall-off: on release the pitch sweeps down by Depth over Time while the note rings, then the note releases and the bend snaps back to centre. The note-off is delayed to the end of the fall.

Param	Range	Default	Unit
Depth	0 - 1	0.5	-
Time	10 - 2000	300	ms

MidiVibrato 1 input

A pitch-bend LFO that fades in after an Onset delay while any note is held - the delayed, played-in vibrato of a string or voice rather than an instant wobble. Rate in Hz, Depth as bend amount.

Param	Range	Default	Unit
Rate	0.1 - 12	5	Hz
Depth	0 - 1	0.15	-
Onset	0 - 2000	300	ms

Harmony

37 modules

Chord 1 input

Turns each incoming note into a chord (165 types) with inversion, spread, octave doubling and strum.

Param	Range	Default
Type	Unison · Octave · 5th (Power) · 5th + Oct · Tritone · Major · Minor · Diminished · Augmented · Major (1st inv) · Major (2nd inv)	
Inversion	0 - 4	0
Spread	0 - 2	0
Octave x2	0 - 2	0
Transpose	-24 - 24	0
Strum	0 - 200	0

Scale 1 input

Quantizes incoming notes to the nearest degree of the chosen root and scale.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pentatonic	

Transpose 1 input

Shifts every note by a fixed semitone amount plus a modulatable offset.

Param	Range	Default	Unit
Semitones	-24 - 24	0	st
Mod Amt	-24 - 24	0	st

Harmonize 1 input

Adds up to three fixed-interval harmony voices to each note.

Param	Range	Default	Unit
Interval 1	-24 - 24	7	st
Interval 2	-24 - 24	0	st
Interval 3	-24 - 24	0	st

Drone 1 input

Doubles every note with a fixed pedal root.

Param	Range	Default	Unit
Root	0 - 127	36	-

MidiCluster 1 input

Adds symmetric tone-cluster voices above and below each note.

Param	Range	Default	Unit
Width	0 - 4	1	—
Spread	1 - 4	1	st

Glissando 1 input

Fills a stepwise run between consecutive notes.

Param	Range	Default	Unit
Rate	5 - 200	40	ms

ChordMem 1 input

Each key plays a stored chord shape (semitone intervals from node config).

ChordProg 1 input

Each note triggers the next chord in a stored root progression (node config).

ChordVoicing 1 input

Re-voices each note into a jazz shape (open / shell / quartal / spread).

Param	Range	Default	Unit
Voicing	Open · Shell · Quartal · Spread	—	—

Octave 1 input

Doubles each note up and/or down by whole octaves.

Param	Range	Default	Unit
Up	0 - 3	1	—
Down	0 - 3	0	—

MidiFold 1 input

Folds out-of-range notes back inside a Low..High window.

Param	Range	Default	Unit
Low Note	0 - 127	48	—
High Note	0 - 127	72	—

Invert 1 input

Mirrors pitch around a pivot note.

Param	Range	Default	Unit
Pivot	0 - 127	60	-

Diatonic 1 input

Transposes by scale DEGREES within Root/Scale (a third, a fifth...) instead of by raw semitones like Transpose - the shift always lands back in key. Mod Amt adds a modulatable degree offset.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · P	
Degrees	-7 - 7	2
Mod Amt	-7 - 7	0

DiatonicChord 1 input

Auto-harmoniser: builds the in-key chord on every note by stacking diatonic thirds (triad, 7th...) along Root/Scale, so single-finger melodies become correct chords in the key.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pen	
Notes	2 - 4	3

NeoRiemann 1 input

Neo-Riemannian triad transforms: builds a major/minor triad on each note and applies P (parallel), L (leading-tone) or R (relative) - the maximally-smooth chord moves of late-Romantic and film harmony, each sharing two common tones with the input.

Param	Range	Default	Unit
Quality	Major · Minor		-
Transform	None · P · L · R		-

Counterpoint 1 input

Adds a second voice in contrary motion - when the melody rises the counter voice falls and vice-versa - quantised to Root/Scale. Interval sets the starting separation.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian ·	
Interval	1 - 24	7

Bassline 1 input

Follows the chord with a mono bass voice: the lowest held note dropped Octaves down, retriggered whenever the lowest note changes. Passes the original notes through and adds the bass beneath them.

Param	Range	Default	Unit
Octaves	1 - 3	1	-

Param	Range	Default	Unit
Velocity	1 - 127	100	-

VoiceLead 1 input

Smooth voice leading: places each note's pitch class in the octave nearest the previous output, so a line or chord progression moves by the smallest interval. Range caps the leap; past it the original octave is kept.

Param	Range	Default	Unit
Range	1 - 12	7	st

AutoInvert 1 input

Cycles the inversion of each struck chord: every trigger rotates which of the lowest notes jump an octave (by Rotate), so a repeated chord keeps climbing through its inversions - movement ChordVoicing's static shapes don't give.

Param	Range	Default	Unit
Rotate	1 - 4	1	-

ChordLock 1 input

Quantizes every note to the nearest tone of a fixed chord (Root + Type), tighter than Scale's scale-quantize - locks an improvisation to a chord's notes.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Type	Unison · Octave · 5th (Power) · 5th + Oct · Tritone · Major · Minor · Diminished · Augmented · Major (1st inv) · Major (2nd	

GhostNote 1 input

Ghost note: doubles every note with a quiet copy a set interval below (default an octave), at a fraction of the original velocity - the subliminal low reinforcement drummers and bassists use to thicken a line.

Param	Range	Default	Unit
Interval	1 - 24	12	st
GhostLevel	0.05 - 1	0.4	-

MirrorPivot 1 input

Mirror pivot: reflects every note about a fixed pivot pitch (out = 2*Pivot - note), turning rising lines into falling ones and inverting every interval around the chosen axis - melodic inversion with a movable mirror.

Param	Range	Default	Unit
Pivot	0 - 127	60	-

OctaveClamp 1 input

Octave clamp: folds every incoming note into the single octave starting at Center by collapsing its pitch class, flattening wide-ranging input into one register - a pitch-class monitor or tight-range re-voicer.

Param	Range	Default	Unit
Center	0 - 116	48	—

SubHarmonicAdd 1 input

Sub-harmonic add: reinforces every note with undertone voices an octave (Sub1) and an octave-plus-fifth (Sub2) below at adjustable levels - the subharmonic weight of a Trautonium or organ bourdon.

Param	Range	Default	Unit
Sub1	0 - 1	0.5	—
Sub2	0 - 1	0.25	—

OvertoneStack 1 input

Overtone stack: layers quieter harmonic-series voices (octave, octave+fifth, two octaves) above each note with a geometric Rolloff, thickening a single note toward an organ-like harmonic chord.

Param	Range	Default	Unit
Rolloff	0.1 - 1	0.5	—

PitchClassGate 1 input

Pitch-class gate: passes only notes whose pitch class belongs to the chosen Root and Mode set (chromatic, major, minor, pentatonic or whole-tone), filtering a stream down to a key without transposing anything.

Param	Range	Default	Unit
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	—	
Mode	Chromatic · Major · Minor · Pentatonic · WholeTone	—	

WholeToneSnap 1 input

Whole-tone snap: quantizes every note to the nearest degree of the six-note whole-tone scale from Root, dissolving tonality into the dreamy, rootless ambiguity that scale is famous for.

Param	Range	Default	Unit
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	—	

ReflectOctave 1 input

Reflect octave: mirrors each note's pitch class about a pivot while keeping it in its own octave, inverting the melodic contour within every register at once - octave-preserving interval inversion.

Param	Range	Default	Unit
Pivot	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	—	

NegativeHarmony 1 input

Negative harmony: reflects every note about the axis halfway between the tonic and dominant (Ernst Levy's mirror), swapping major for minor and turning a progression into its tonal shadow while keeping the same harmonic function.

Param	Range	Default	Unit
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	—	—

TwelveToneRow 1 input

Twelve-tone row: enforces Schoenberg's serial rule - no pitch class may sound again until all twelve have been played - dropping early repeats so the stream is forced into complete chromatic rows.

Param	Range	Default	Unit
On	0 - 1	1	—

ConsonanceGate 1 input

Consonance gate: passes a note only when its interval from the previous one is consonant (unison, third, fourth, fifth, sixth or octave), filtering a chromatic line down to smooth, dissonance-free melodic motion.

Param	Range	Default	Unit
On	0 - 1	1	—

GravityQuantize 1 input

Gravity quantize: pulls every note toward a center pitch by an adjustable strength, as if the keyboard had a gravitational well - gentle settings nudge stray notes inward, strong settings collapse a whole part toward one tone.

Param	Range	Default	Unit
Center	0 - 127	60	—
Strength	0 - 1	0.3	—

HarmonicQuantize 1 input

Harmonic quantize: re-pitches each note to the nearest member of the natural harmonic series above a root, snapping a chromatic part onto the pure, stretched-then-compressed intervals of the overtone series.

Param	Range	Default	Unit
Root	0 - 60	24	—

SubharmonicShift 1 input

Subharmonic shift: drops each note to the nearest undertone (subharmonic) of a ceiling pitch, the mirror of the harmonic series, building dark, minor-tilted lines beneath an upper anchor note.

Param	Range	Default	Unit
Ceiling	60 - 120	96	—

RangeCompress 1 input

Range compress: linearly squeezes the played pitch range toward a center note, narrowing a wide-ranging part into a tighter register (or, near ratio 1, leaving it alone) - a melodic-ambitus compressor.

Param	Range	Default	Unit
Center	0 - 127	60	-
Ratio	0.1 - 1	0.6	-

RangeExpand 1 input

Range expand: stretches the played pitch range away from a center note (the inverse of range-compress), exaggerating a cramped melody into a wider, more dramatic ambitus; near ratio 1 it leaves the part alone.

Param	Range	Default	Unit
Center	0 - 127	60	-
Ratio	1 - 3	1.5	-

Probability

48 modules

Chance 1 input

Probabilistic gate: lets each note through with the set probability.

Param	Range	Default	Unit
Probability	0 - 1	0.8	-
Mod Amt	-1 - 1	0	-

PrimeGate 1 input

Number-theory gate: counts incoming notes and lets one through only when its ordinal is a prime number (2,3,5,7,11,...), so the texture thins out unpredictably as primes spread; Invert keeps the composites instead.

Param	Range	Default	Unit
Invert	0 - 1	0	-

ThueMorseGate 1 input

Thue-Morse gate: passes notes where the cube-free Thue-Morse sequence (the bit-parity of the note count) is 0, an aperiodic-yet-structured pattern that never settles into a short loop; Invert flips which half plays.

Param	Range	Default	Unit
Invert	0 - 1	0	-

DigitGate 1 input

Digit-sum gate: passes a note unless the decimal digit-sum of its count is divisible by Modulo, carving a self-similar arithmetic rhythm out of a steady stream; Invert keeps only the divisible ones.

Param	Range	Default	Unit
Modulo	2 - 9	3	-
Invert	0 - 1	0	-

CollatzGate 1 input

Collatz gate: runs the $3n+1$ process on each note's count and passes it when the number of steps to reach 1 is even - an erratic but deterministic gate straight from the unsolved Collatz conjecture; Invert takes the odd-length ones.

Param	Range	Default	Unit
Invert	0 - 1	0	-

GoldenGate 1 input

Golden gate: passes notes by a golden-ratio low-discrepancy sequence against Density, so the kept notes spread far more evenly through time than random chance ever would - a deterministic, clump-free thinning.

Param	Range	Default	Unit
Density	0 - 1	0.5	-

FibGate 1 input

Fibonacci gate: passes a note unless the Fibonacci number of its ordinal is divisible by Modulo, carving a golden, self-similar rhythm out of a steady stream; Invert keeps the divisible ones.

Param	Range	Default	Unit
Modulo	2 - 12	5	-
Invert	0 - 1	0	-

RandomGateWalk 1 input

Random-gate walk: the pass probability itself drifts on a bounded random walk, so the music breathes between dense and sparse passages instead of thinning uniformly like a fixed-chance gate.

Param	Range	Default	Unit
Step	0 - 1	0.3	-

IntervalGate 1 input

Interval gate: passes a note only if its leap from the previous kept note lies between Min and Max semitones, filtering a line down to small steps or to wide jumps - a melodic-contour sieve.

Param	Range	Default	Unit
Min	0 - 24	0	st
Max	0 - 24	7	st

VelRandomGate 1 input

Velocity-random gate: passes each note with a probability that rises with its velocity, so accented hits survive while ghost notes thin out - dynamics-weighted probabilistic gating.

Param	Range	Default	Unit
Bias	0 - 1	0.7	-

MoebiusGate 1 input

Moebius gate: passes a note only when its count is squarefree (the Moebius function is non-zero), dropping any note whose ordinal is divisible by a perfect square - a sieve-flavoured aperiodic gate; Invert keeps the square-divisible ones.

Param	Range	Default	Unit
Invert	0 - 1	0	-

SquareGate 1 input

Square gate: passes note-ons only on perfect-square counts (1,4,9,16,25,...), so notes get steadily sparser as the gaps between squares widen; Invert plays everything except the squares.

Param	Range	Default	Unit
Invert	0 - 1	0	-

TriangularGate 1 input

Triangular gate: passes note-ons on triangular-number counts (1,3,6,10,15,...), thinning the stream on the handshake-number sequence; Invert keeps the non-triangular notes.

Param	Range	Default	Unit
Invert	0 - 1	0	-

AbundantGate 1 input

Abundant gate: passes note-ons only on abundant counts - numbers whose proper divisors sum to more than themselves (12, 18, 20, 24...) - thinning the stream onto the divisor-rich integers; Invert keeps the deficient ones.

Param	Range	Default	Unit
Invert	0 - 1	0	-

HappyGate 1 input

Happy gate: passes notes on happy-number counts - those whose repeated sum-of-squared-digits reaches 1 - and drops the sad ones that fall into the 4-16-37 cycle; Invert flips which set plays.

Param	Range	Default	Unit
Invert	0 - 1	0	-

PrimePowerGate 1 input

Prime-power gate: passes note-ons only when the count is a power of a single prime (2,3,4,5,7,8,9,...), a sparse self-similar pattern from analytic number theory; Invert keeps everything else.

Param	Range	Default	Unit
Invert	0 - 1	0	-

LucasGate 1 input

Lucas gate: passes note-ons only on Lucas-number counts (2,1,3,4,7,11,18,...), the Fibonacci companion sequence, thinning the stream onto golden-ratio-spaced ordinals; Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

CatalanGate 1 input

Catalan gate: passes note-ons only on Catalan-number counts (1,2,5,14,42,...), the combinatorial sequence counting trees and bracketings, which thins out fast as the values explode; Invert keeps the others.

Param	Range	Default	Unit
Invert	0 - 1	0	-

PalindromeGate 1 input

Palindrome gate: passes note-ons only on counts that read the same forwards and backwards (1,2,...,9,11,22,...,101,...), an irregular, self-mirroring thinning; Invert keeps the non-palindromes.

Param	Range	Default	Unit
Invert	0 - 1	0	-

HighlyCompositeGate 1 input

Highly-composite gate: passes note-ons only on counts that set a new record for number of divisors (1,2,4,6,12,24,36,48,60,...), the maximally-factorable integers; a rare, widening-gap accent pulse. Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

ProbabilityRamp 1 input

Probability ramp: the pass chance sweeps from Start to End across a repeating cycle of notes, so phrases can fade in (sparse to dense) or thin out (dense to sparse) on a loop - an evolving probabilistic gate.

Param	Range	Default	Unit
Length	2 - 32	8	-
Start	0 - 1	0.2	-
End	0 - 1	1	-

AntiRepeatGate 1 input

Anti-repeat gate: drops a note when it repeats the immediately-previous pitch, forcing the line to keep moving instead of hammering one note - an automatic de-stutter for melodic variety.

Param	Range	Default	Unit
On	0 - 1	1	-

ProbabilityByPitch 1 input

Probability by pitch: the chance a note passes rises or falls with its keyboard position, so you can thin out the bass while keeping the top (or the reverse) - register-weighted probabilistic gating.

Param	Range	Default	Unit
Base	0 - 1	0.7	-
Slope	-1 - 1	0.3	-

PrimeIntervalGate 1 input

Prime-interval gate: passes a note only when its leap from the previous one spans a prime number of semitones (2,3,5,7,11,...), favouring the seconds, thirds, fourths and fifths while blocking octaves and tritones - a number-theoretic melodic filter.

Param	Range	Default	Unit
On	0 - 1	1	-

PerrinGate 1 input

Perrin gate: passes note-ons on Perrin-sequence counts (3,2,5,5,7,10,12,...; $P(n)=P(n-2)+P(n-3)$), famous because n almost always divides $P(n)$ exactly when n is prime; Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

PartitionGate 1 input

Partition gate: passes note-ons on integer-partition counts $p(k)$ (1,2,3,5,7,11,15,22,30,...), the number of ways to write k as a sum, which thins out as it climbs; Invert keeps the others.

Param	Range	Default	Unit
Invert	0 - 1	0	-

TribonacciGate 1 input

Tribonacci gate: passes note-ons on Tribonacci-sequence counts (1,2,4,7,13,24,44,...; each the sum of the previous three), spreading wider than Fibonacci; Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

JacobsthalGate 1 input

Jacobsthal gate: passes note-ons on Jacobsthal-number counts (1,3,5,11,21,43,...; $J(n)=J(n-1)+2J(n-2)$), a Fibonacci-like sequence whose ratio tends to 2; Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

PellGate 1 input

Pell gate: passes note-ons on Pell-number counts (1,2,5,12,29,70,...; $P(n)=2P(n-1)+P(n-2)$), the silver-ratio sequence behind the best rational approximations to root-2; Invert keeps the others.

Param	Range	Default	Unit
Invert	0 - 1	0	-

CoprimeGate 1 input

Coprime gate: passes a note only when its count shares no common factor with the modulus ($\gcd = 1$), thinning the stream onto the totatives of the modulus - a number-theoretic sieve; Invert keeps the non-coprime counts.

Param	Range	Default	Unit
Modulus	2 - 16	6	-
Invert	0 - 1	0	-

ContraryGate 1 input

Contrary gate: passes a note only if it moves in the opposite direction to the previous melodic step, forcing the line into constant zig-zag contrary motion and filtering out runs that keep climbing or falling.

Param	Range	Default	Unit
On	0 - 1	1	-

WythoffGate 1 input

Wythoff gate: passes note-ons on the lower-Wythoff counts $\text{floor}(n \cdot \phi)$ (1,3,4,6,8,9,11,...), the golden-ratio Beatty sequence that, with its complement, partitions the integers; Invert keeps the upper-Wythoff notes.

Param	Range	Default	Unit
Invert	0 - 1	0	-

SemiprimeGate 1 input

Semiprime gate: passes note-ons on semiprime counts - numbers that are the product of exactly two primes (4,6,9,10,14,15,21,...) - a moderately-sparse, multiplicatively-defined pattern; Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

PentagonalGate 1 input

Pentagonal gate: passes note-ons on pentagonal-number counts (1,5,12,22,35,...; $k(3k-1)/2$), the figurate numbers from Euler's pentagonal-number theorem; Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

PronicGate 1 input

Pronic gate: passes note-ons on pronic (oblong) counts $n(n+1)$: 2,6,12,20,30,..., the products of consecutive integers; Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

MersenneGate 1 input

Mersenne gate: passes note-ons on Mersenne-number counts (one less than a power of two: 1,3,7,15,31,63,127), the all-ones binary numbers; their exponential spacing thins the stream fast. Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

TwinPrimeGate 1 input

Twin-prime gate: passes note-ons on counts belonging to a twin-prime pair (a prime with another prime two away: 3,5,7,11,13,17,19,...), thinning onto the famously-clustered twin primes; Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

SophieGermainGate 1 input

Sophie-Germain gate: passes note-ons on Sophie-Germain-prime counts - primes p where $2p+1$ is also prime (2,3,5,11,23,...), the primes behind safe-prime cryptography; Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

RecamanGate 1 input

Recaman gate: passes note-ons on counts that appear in Recaman's sequence (jump back by n if new and positive, else forward), the famously erratic jumping sequence; its darting coverage thins the stream unpredictably. Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

UlamGate 1 input

Ulam gate: passes note-ons on Ulam-number counts (each the smallest integer that is a sum of two earlier terms in exactly one way: 1,2,3,4,6,8,11,...), a mysteriously near-periodic set; Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

TetrahedralGate 1 input

Tetrahedral gate: passes note-ons on tetrahedral-number counts $n(n+1)(n+2)/6$ (1,4,10,20,35,56,...), the 3-D figurate numbers counting stacked spheres; their cubic spacing thins fast. Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

NarcissisticGate 1 input

Narcissistic gate: passes note-ons on Armstrong (narcissistic) counts that equal the sum of their own digits each raised to the number-of-digits power (1,153,370,371,407,...), a rare self-referential set; Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

PolygonalGate 1 input

Polygonal gate: passes note-ons on s-gonal figurate-number counts for any chosen number of Sides - triangular at 3, square at 4, pentagonal at 5, and so on - one knob covering the whole figurate family; Invert keeps the rest.

Param	Range	Default	Unit
Sides	3 - 12	5	-
Invert	0 - 1	0	-

SmoothNumberGate 1 input

Smooth-number gate: passes note-ons whose count's largest prime factor stays at or below Bound (a B-smooth number, the easy-to-factor integers behind sieve algorithms); raising Bound opens the gate on more counts. Invert keeps the rest.

Param	Range	Default	Unit
Bound	2 - 50	7	-
Invert	0 - 1	0	-

RoughNumberGate 1 input

Rough-number gate: passes note-ons whose count's smallest prime factor is at least Bound (a B-rough number with no small factors), the complement of smoothness; raising Bound makes the gate fire only on prime-like counts. Invert keeps the rest.

Param	Range	Default	Unit
Bound	2 - 30	5	-
Invert	0 - 1	0	-

KeithNumberGate 1 input

Keith-number gate: passes note-ons on Keith (repfigit) counts - numbers that reappear in the Fibonacci-like sequence seeded by their own digits (14,19,28,47,...) - an extremely rare self-referential set; Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

PracticalNumberGate 1 input

Practical-number gate: passes note-ons on practical counts, where every smaller integer is a sum of distinct divisors of the count (1,2,4,6,8,12,...; all powers of two and more), a divisor-rich set close to the abundant numbers; Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

UndulatingGate 1 input

Undulating gate: passes note-ons on undulating counts whose digits zig-zag high-low-high (121,132,231,1212,...), a wave-like digit pattern; Invert keeps the rest.

Param	Range	Default	Unit
Invert	0 - 1	0	-

Routing

18 modules

Split 1 input

Keyboard split: notes below the split point transpose by Low, the rest by High.

Param	Range	Default	Unit
Split	0 - 127	60	–
Low	-24 - 24	0	st
High	-24 - 24	0	st

Transformer 1 input

Rule: notes within a range are transposed and velocity-scaled.

Param	Range	Default	Unit
Low	0 - 127	0	–
High	0 - 127	127	–
Transpose	-24 - 24	0	st
VelScale	0 - 2	1	–

Merge 2 inputs

Passes both note-stream inputs through, merged into one stream.

NoteFilter 1 input

Passes only notes inside a key and velocity window.

Param	Range	Default	Unit
Low Note	0 - 127	0	–
High Note	0 - 127	127	–
Low Vel	1 - 127	1	–
High Vel	1 - 127	127	–

Midilatch 1 input

Holds notes after release until the next note arrives.

Channel 1 input

Reassigns notes to a fixed MIDI channel.

Param	Range	Default	Unit
Channel	1 - 16	1	–

FixedNote 1 input

Forces every note to a single fixed pitch (drum / trigger).

Param	Range	Default	Unit
Note	0 - 127	36	-

KeySwitch 1 input

Splits the keyboard at a note; one zone passes, the other is filtered.

Param	Range	Default	Unit
Split	0 - 127	60	-
Low Zone	High passes · Low passes		-

NoteLimit 1 input

Polyphony cap: never lets more than Max notes sound at once, stealing the oldest (FIFO note-off) when a new note would exceed the limit - tames dense chords and runaway generators.

Param	Range	Default	Unit
Max	1 - 16	4	-

PolyChan 1 input

Round-robins each new note to the next of Voices MIDI channels from Base, spreading a polyphonic part across a multitimbral rig - one note per channel, the classic voice-allocation split (distinct from MPEConvert's per-note MPE channels).

Param	Range	Default	Unit
Voices	1 - 16	4	-
Base	1 - 16	1	-

Thin 1 input

Note-density limiter: drops any note-on arriving within MinGap ms of the last one passed, decluttering a busy stream or fast repeats - distinct from NoteLimit, which caps simultaneous polyphony. A dropped note's off is dropped too.

Param	Range	Default	Unit
MinGap	1 - 500	30	ms

MaxPolyphony 1 input

Max polyphony: caps how many notes may sound at once, dropping any note-on past the limit until a voice frees up - a voice-stealing-free polyphony clamp for thinning dense chords.

Param	Range	Default	Unit
Voices	1 - 32	4	-

FoldToRange 1 input

Fold to range: shifts every note up or down by whole octaves until it sits inside the [Low,High] window, packing a wide-range part into a fixed register while preserving each note's pitch class.

Param	Range	Default	Unit
Low	0 - 127	48	—
High	0 - 127	72	—

WrapAround 1 input

Wrap around: notes that rise above the High edge wrap back to the Low end of the range (and vice versa), a modular pitch fold that keeps a part inside a register while letting runs cycle around.

Param	Range	Default	Unit
Low	0 - 127	48	—
High	0 - 127	72	—

NoteToPitchBend 1 input

Note-to-pitch-bend: emits a pitch-bend proportional to each note's distance from a center pitch, so notes far from the center are pre-bent - a way to map keyboard position onto continuous detune or to drive a synth's bend input from played pitch.

Param	Range	Default	Unit
Center	0 - 127	60	—
Amount	0 - 1	0.5	—

RandomPanCC 1 input

Random-pan CC: fires a random pan (controller 10) value on every note-on, scattering successive notes across the stereo field for an automatic, ever-shifting spatial spread; Spread sets how wide the scatter.

Param	Range	Default	Unit
Spread	0 - 1	0.8	—

KeyboardSplit3 1 input

Three-zone split: divides the keyboard at two split points into low, mid and high zones, each transposed by its own amount - a quick way to layer a bassline, a pad and a lead from one part across three regions.

Param	Range	Default	Unit
Split1	0 - 127	48	—
Split2	0 - 127	72	—
Low	-24 - 24	-12	st
Mid	-24 - 24	0	st
High	-24 - 24	12	st

ChannelByPitch 1input

Channel-by-pitch: routes notes to different MIDI channels by register, splitting the keyboard into Zones equal bands so a single part can drive several multitimbral instruments stacked across the range.

Param	Range	Default	Unit
Zones	2 - 16	4	-

Sequencing

46 modules

Arp 1 input

Arpeggiator: cycles held notes through a pattern at a synced rate with gate, octaves, swing and step count. Latch holds the chord so it keeps arpeggiating after the keys are released (the next key starts a new chord).

Param	Range	Default	Unit
Mode	Up · Down · Up/Down · Down/Up · Converge · Diverge · As Played · Random · Chord		–
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		–
Gate	0 – 2	0.5	–
Octaves	1 – 4	1	–
Swing	0 – 0.75	0	–
Steps	1 – 16	16	–
Latch	0 – 1	0	–

SeqEuclid 1 input

Euclidean rhythm gate: spreads Pulses evenly across Steps (rotatable) at a synced rate.

Param	Range	Default	Unit
Pulses	1 – 16	5	–
Steps	1 – 16	8	–
Rotate	0 – 15	0	–
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		–
Gate	0 – 2	0.5	–

MidiGameOfLife 1 input

Conway's Game of Life MIDI sequencer: an 8x8 cell grid evolves under the B3/S23 rule; each clock step reads the next column and plays a note for every live cell (row maps to a scale degree from Root/Scale), and a full 8-step sweep advances one generation. Gliders, blinkers and still-lives become ever-shifting melodic and harmonic patterns. Density sets the random fill when the grid (re)seeds. The MIDI-note counterpart to the GameOfLife CV block.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · P	
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	
Gate	0 – 2	0.5
Density	0 – 1	0.35

MidiBrianBrain 1 input

Brian's Brain MIDI sequencer: an 8x8 three-state grid (ready / firing / dying) evolves under Brian's Brain rules - a ready cell ignites only with exactly two firing neighbours, firing cells pass to dying, dying cells reset. Each clock step plays a note for every firing cell in the next column (row maps to a scale degree from Root/Scale); a full 8-step sweep advances one generation. The refractory state keeps it restless and sparkly where Game of Life settles. Density sets the random fill. The MIDI-note counterpart to the BrianBrain CV block.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · P	
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	
Gate	0 - 2	0.5
Density	0 - 1	0.3

Midilangton 1 input

Langton's Ant MIDI melody: a single turmite walks an 8x8 grid - turning right on a white cell, left on a black one, flipping each cell as it leaves, then stepping forward. Each clock advances the ant Steps cells and plays one note for its current row (mapped to a scale degree from Root/Scale), so the melody scribbles chaotically then locks into the famous periodic 'highway'. Emergent order from one trivial rule - a single moving agent, not a population.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pen	
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	
Gate	0 - 2	0.5
Steps	1 - 16	1

Swing 1 input

Pushes off-beat eighth-notes later for a swing/shuffle feel.

Param	Range	Default	Unit
Amount	0 - 0.75	0.3	-

PatternGate 1 input

Gates note-ons through a 16-step on/off pattern (bitmask).

Param	Range	Default	Unit
Pattern	0 - 65535	21845	-
Steps	1 - 16	16	-

NoteOffDelay 1 input

Extends each note's gate by delaying its note-off.

Param	Range	Default	Unit
Delay	0 - 2000	200	ms

Ricochet 1 input

Accelerating bouncing retriggers with decaying velocity.

Param	Range	Default	Unit
Bounces	1 - 16	6	-
Decay	0.3 - 0.98	0.7	-
Start	10 - 500	100	ms

StepSeqMidi 1 input

Melodic step sequencer - semitone offsets (node config) clock the held root.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		—
Gate	0.05 - 1	0.5	—
Velocity	1 - 127	100	—

Echo 1 input

MIDI delay: repeats notes at a synced time with velocity feedback decay.

Param	Range	Default	Unit
Time	1 - 2000	250	ms
Feedback	0 - 0.95	0.5	—
Repeats	1 - 8	3	—

Flam 1 input

Precedes every note with a quiet grace note, delaying the main hit so the grace leads it - the classic drum flam.

Param	Range	Default	Unit
Time	1 - 200	30	ms
Grace Vel	0 - 1	0.5	—

Roll 1 input

Drum roll / buzz: retriggers every held note at Rate for as long as it is held, each hit gated to a fraction of the interval.

Param	Range	Default	Unit
Rate	1 - 50	12	Hz
Gate	0.05 - 0.95	0.5	—

Trill 1 input

Keyboard trill: while a note is held, alternates rapidly between it and the note Interval semitones above, at Rate.

Param	Range	Default	Unit
Rate	1 - 50	10	Hz
Interval	1 - 12	2	st
Gate	0.05 - 0.95	0.5	—

Mordent 1 input

One-shot ornament: each struck note plays principal -> auxiliary -> principal at the attack, then sustains, the single-flick counterpart to the continuous Trill.

Param	Range	Default	Unit
Time	5 - 150	40	ms
Interval	1 - 12	2	st

Param	Range	Default	Unit
Direction	Upper · Lower		–

Turn 1 input

Gruppetto turn: a four-step ornament at the attack - upper auxiliary, principal, lower auxiliary, then the principal held. Time sets each step, Interval the neighbour distance.

Param	Range	Default	Unit
Time	5 - 150	35	ms
Interval	1 - 12	2	st

Appoggiatura 1 input

Leaning grace note: each struck note first sounds a long auxiliary (Interval away) that takes Time from the principal, which then sustains - the expressive long grace, unlike the quick Flam or Mordent.

Param	Range	Default	Unit
Time	20 - 400	120	ms
Interval	1 - 12	2	st
Direction	Upper · Lower		–

Strum 1 input

Spreads a chord's notes over time, up or down, like a guitar strum.

Param	Range	Default	Unit
Spread	0 - 200	20	ms
Direction	Up · Down		–

MidiQuantize 1 input

Snaps note starts to a synced grid by a strength amount.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		–
Strength	0 - 1	1	–

SeqRatchet 1 input

Retriggers each note Count times within a synced step.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		–
Count	1 - 8	4	–
Gate	0.05 - 1	0.9	–

NoteLength 1 input

Forces a fixed note duration (gate length).

Param	Range	Default	Unit
Length	1 - 2000	200	ms

Cascade 1 input

Each struck note fans out into a timed run of Steps notes climbing (or falling) the scale from it - an arpeggiated flourish fired per key, unlike Arp which cycles the whole held chord.

Param	Range	Default
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	
Scale	Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian ·	
Steps	2 - 8	4
Time	5 - 200	50
Direction	Up · Down	

MidiTuring 1 input

Turing-machine sequencer (Music Thing style): a circular shift register clocks once per step and the held notes fire when the read bit is 1. Change is the probability the recycled bit flips - 0 locks an evolving loop, 1 fully randomises, between = slowly mutating melodies.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	-	-
Length	2 - 16	8	-
Change	0 - 1	0.15	-
Gate	0.05 - 1	0.5	-

Polymeter 1 input

Two step lanes of different lengths (LenA, LenB) clock together over the held notes, phasing against each other so the combined pattern only repeats after lcm(LenA,LenB) steps - instant polymetric movement from one chord.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	-	-
Len A	1 - 16	3	-
Len B	1 - 16	4	-
Gate	0.05 - 1	0.5	-

MidiTranceGate 1 input

Chops the held notes with a 16-step on/off Pattern at a synced Rate: every 'on' step retriggers all held notes, gated to a fraction of the step. The rhythmic gate behind the trance sound - pattern-driven, where Roll is a fixed pulse.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	-	-
Pattern	0 - 65535	21845	-
Steps	1 - 16	16	-
Gate	0.05 - 1	0.5	-

StepArp 1 input

Cthulhu-style step arpeggiator: walks the held chord strictly upward across Octaves octaves at a synced Rate, accenting the first step of each cycle - the rolling, octave-jumping arp of Xfer Cthulhu's sequencer. Distinct from Arp's mode-cycling: this climbs the chord in order.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		–
Gate	0.05 – 1	0.5	–
Octaves	1 – 4	2	–
Accent	1 – 127	110	–

ChordArp 1 input

Cthulhu-style chord arpeggiator: each key builds a full chord (Type) and the result is arpeggiated upward across Octaves at a synced Rate - one finger plays a moving arpeggio of the whole chord. Releasing the key stops its run.

Param	Range	Default
Type	Unison · Octave · 5th (Power) · 5th + Oct · Tritone · Major · Minor · Diminished · Augmented · Major (1st inv) · Major (2n	
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	
Gate	0.05 – 1	0.5
Octaves	1 – 4	1

OctaveArp 1 input

Cthulhu's arp octave lane: arpeggiates the held chord while a per-step octave-offset Pattern leaps the line up and down by whole octaves (alternating, staircase, bounce...) instead of climbing monotonically like StepArp.

Param	Range	Default	Unit
Pattern	0 – 7	1	–
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		–
Gate	0.05 – 1	0.5	–

GateArp 1 input

Cthulhu's arp gate lane: walks the held chord upward across Octaves while the note length cycles a per-step staccato/legato pattern, so the rhythm breathes - short stabs then sustained notes. Gate scales the pattern (>1 overlaps into the next step).

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		–
Gate	0.25 – 2	1	–
Octaves	1 – 4	1	–

VelArp 1 input

Cthulhu's arp velocity lane as a ramp: arpeggiates the held chord while velocity sweeps from From to To across Steps then resets, so every pass swells or fades without a per-step editor.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		–
Gate	0.05 – 1	0.5	–
From	1 – 127	40	–

Param	Range	Default	Unit
To	1 - 127	120	-
Steps	1 - 32	8	-

TransArp 1 input

Cthulhu's arp transpose lane: arpeggiates the held chord upward across Octaves while a per-step semitone Pattern (fifth pedals, triad arps, scalar runs) is added on top, so the line outlines a melodic contour, not just the chord tones.

Param	Range	Default	Unit
Pattern	0 - 7	2	-
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	-	-
Gate	0.05 - 1	0.5	-
Octaves	1 - 4	1	-

RatchetArp 1 input

Cthulhu's arp repeat lane: walks the held chord upward across Octaves while a per-step pattern retriggers selected steps multiple times (capped by Max), packing rolls into the line. Unlike SeqRatchet's fixed count, the repeat count varies per step.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	-	-
Gate	0.05 - 1	0.5	-
Octaves	1 - 4	1	-
Max	1 - 8	4	-

SkipArp 1 input

Cthulhu's arp skip lane: arpeggiates the held chord upward across Octaves, but each step has a Skip probability of being dropped to a rest, thinning the pattern differently every cycle for generative movement.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	-	-
Gate	0.05 - 1	0.5	-
Octaves	1 - 4	1	-
Skip	0 - 0.95	0.3	-

Drunk 1 input

Random-walk melodic sequencer: each synced Rate step nudges the pitch by a random +/-Step semitones, bounded within +/-Range of the lowest held note - Brownian melody anchored to the chord, unlike MidiRandom's memoryless picks.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T	-	-
Gate	0.05 - 1	0.5	-
Step	1 - 7	2	st
Range	1 - 24	12	st

EuclidMel 1input

Euclidean melody: spreads Pulses evenly across Steps (Bjorklund) and, on each hit, advances a cursor through the held chord - the rhythm is Euclidean and the pitch walks the chord, unlike SeqEuclid which stabs all held notes per pulse.

Param	Range	Default	Unit
Pulses	1 - 16	5	-
Steps	1 - 16	8	-
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		-
Gate	0.05 - 1	0.5	-

Spiral 1input

Transposing MIDI delay: each repeat is delayed by Time and shifted by Interval semitones with velocity decaying by Feedback, so one note spins off a climbing or falling staircase of echoes - unlike Echo, which repeats the same pitch.

Param	Range	Default	Unit
Time	1 - 1000	120	ms
Interval	-12 - 12	4	st
Feedback	0 - 0.95	0.7	-
Repeats	1 - 12	5	-

Tuplet 1input

Snaps note starts to a Triplets-per-beat grid (3 = triplets, 5 = quintuplets) by Strength, for a triplet or odd-tuplet feel where MidiQuantize only snaps to the straight grid.

Param	Range	Default	Unit
Tuplets	2 - 9	3	-
Strength	0 - 1	1	-

Shift 1input

Delays every note by Delay ms (groove nudge / lay-back), pushing note-ons and offs back by the same amount so durations are preserved - the per-track timing offset of a groove sequencer.

Param	Range	Default	Unit
Delay	0 - 200	20	ms

MidiMarkov 1input

Learns the note-to-note transition statistics of what you play (a 12x12 pitch-class chain) and, on each synced step, walks the chain to improvise a new line in the same style. Memory fades old transitions.

Param	Range	Default	Unit
Rate	1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T		-
Gate	0.05 - 1	0.5	-
Memory	0 - 1	0.2	-

PolyrhythmGate 1 input

Polyrhythm gate: passes notes that land on a multiple of either count A or count B, overlaying two pulse grids (e.g. 3 against 4) into an interlocking cross-rhythm of accents.

Param	Range	Default	Unit
A	1 - 16	3	–
B	1 - 16	4	–

ClockDivideGate 1 input

Clock divider: passes one note-on out of every Divide that arrive and drops the rest, a simple note-rate divider for thinning fast passages or extracting a downbeat pulse.

Param	Range	Default	Unit
Divide	1 - 16	2	–

EuclidComplementGate 1 input

Euclidean complement: passes notes that fall on the rests of a Euclidean rhythm of Pulses-in-Steps, the photo-negative of a Euclid sequencer - an instant interlocking counter-rhythm.

Param	Range	Default	Unit
Pulses	1 - 16	5	–
Steps	1 - 16	8	–

SkipNote 1 input

Skip note: drops every Nth note-on and lets the rest through, the inverse of a clock divider - a quick way to punch regular holes in a run of notes.

Param	Range	Default	Unit
Every	2 - 16	4	–

PaperfoldGate 1 input

Paperfold gate: gates notes by the regular paperfolding (dragon-curve) sequence of the count, a 2-automatic pattern that folds in on itself at every scale; Invert flips the kept and dropped halves.

Param	Range	Default	Unit
Invert	0 - 1	0	–

BurstGate 1 input

Burst gate: lets OnLen consecutive notes through, mutes the next OffLen, and repeats, chopping a steady run into rhythmic bursts and rests - a note-count gate for stutter and ratchet feels.

Param	Range	Default	Unit
OnLen	1 - 16	3	–
OffLen	1 - 16	1	–

BeattyGate 1 input

Beatty gate: passes note-ons on a Beatty sequence $\text{floor}(n \cdot \text{Ratio})$ for any irrational ratio, a Sturmian, maximally-even thinning whose density is set by the ratio; Invert keeps the complement.

Param	Range	Default	Unit
Ratio	1.1 - 3	1.618	-
Invert	0 - 1	0	-

Tuning

3 modules

Microtuner 1 input

Stretched-octave microtuning via per-note pitch detune.

Param	Range	Default	Unit
Stretch	-50 - 50	0	cents/oct

JustTune 1 input

Adaptive just intonation: retunes each note toward its 5-limit JI ratio relative to Key via per-note detune (polyphonic), Amount blending from equal temperament to pure - the beatless thirds and fifths of real harmony.

Param	Range	Default	Unit
Key	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	-	-
Amount	0 - 1	1	-

TuneTable 1 input

Custom microtuning: detunes each note by a per-pitch-class cents offset from the node's 12-value config list (relative to Key) - meantone, Pythagorean, maqam or gamelan, any temperament. Amount scales the table.

Param	Range	Default	Unit
Key	C · C# · D · D# · E · F · F# · G · G# · A · A# · B	-	-
Amount	0 - 1	1	-

Voicing

5 modules

MPEConvert 1 input

Assigns each note its own rotating channel for per-note MPE expression.

Param	Range	Default	Unit
Voices	2 - 15	15	-

Sustain 1 input

Sustain-pedal simulation: while Hold is engaged, note-offs are captured so the notes keep sounding; when Hold drops, every held note releases at once. Hold is a parameter, so a CC, an LFO or automation can play the pedal.

Param	Range	Default	Unit
Hold	Off · On		-

Mono 1 input

Monophonic note priority (last / lowest / highest).

Param	Range	Default	Unit
Priority	Last · Lowest · Highest		-

MidiUnison 1 input

Doubles each note into multiple voices, optionally spread across channels.

Param	Range	Default	Unit
Voices	1 - 4	2	-
Spread	Same chan · Spread chans		-

Slide 1 input

Cthulhu's slide: mono legato glue. Picks one held note by Priority (last/low/high) and emits the new note-on before releasing the old one, so a portamento synth slurs between notes instead of retriggering.

Param	Range	Default	Unit
Priority	Last · Low · High		-